

Origin of the CEEX–PHOKHARA discrepancy in the pion angular distributions

KLOE large-angle scenario, $e^+e^- \rightarrow \pi^+\pi^-\gamma$ at NLO, $F_\pi = 1$, no vacuum polarisation

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Abstract

In the KLOE large-angle (KLOE-LA) scenario the NLO predictions of CEEX and PHOKHARA 10.0 for $e^+e^- \rightarrow \pi^+\pi^-\gamma$ agree in the total cross section and in every charge-even distribution, but disagree in the pion polar angles: the forward–backward asymmetry of the π^+ is $A(\theta^+) = 0.2375$ in CEEX (and 0.2372 in BabaYaga) against 0.2787 in PHOKHARA. We show that the difference is entirely due to two errors in the one-photon ISR×FSR interference at NLO in PHOKHARA (routine `ampL0`, pion branch of the full-NLO mode): a misplaced parenthesis that adds three helicity terms of the tree interference a second time, and a block that keeps the tree interference together with an analytic ISR soft function that the full-NLO mode replaces by the eikonal elsewhere. The second error makes the PHOKHARA asymmetry depend on the unphysical soft-photon cutoff. A point-by-point comparison of the one-photon matrix elements shows that the two errors account for the whole charge-odd difference to 2.5×10^{-5} of the tree. With both corrected, PHOKHARA gives $A(\theta^+) = 0.23724(10)$ averaged over the cutoff and 0.23725(3) with 5×10^9 events, independent of the cutoff; CEEX (0.23725(3)) and BabaYaga (0.23724) agree with it to 3×10^{-5} .

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*Analysis carried out with Claude Code (Anthropic). Repository: https://github.com/jpaltrinieri/CEEX_PHOKHARA_ANALYSIS

1 Summary of findings

1. **The disagreement is reproduced** with a fresh PHOKHARA build and fresh CEEX runs (Section 4): totals agree to 10^{-4} , charge-even shapes agree, and only $\theta^\pm$ and θ_{avg} differ ($\chi^2/\text{ndf} \approx 10^3$). At LO the two codes agree exactly.
2. **The PHOKHARA asymmetry depends on the soft cutoff w** : $A(\theta^+)$ changes by $+0.0156$ per decade of w while σ is constant (Section 6.1).
3. **The whole effect sits in the charge-odd part of the one-photon contribution** (Born + virtual + soft); the hard two-photon emission agrees between the codes (Section 6.3).
4. **Two errors in PHOKHARA explain it point by point** (Sections 6.4 and 7):
 - **H1** – `phokhara_10.0.f:9969-9982`: the factor $(v_f + v_s)$ multiplies only the last of four helicity terms of the tree interference;
 - **H2** – block B5, the ISR one-loop current times the FSR tree (`phokhara_10.0.f:9998-10007`; `ISRcurrent_no_mass_terms.f:134,210`): an extra $\text{Re}[F^*I](1 + S_{\text{ISR}}(w))$.

Subtracting both leaves a per-point residual of 2.5×10^{-5} of the tree.

5. **With both corrected**, $A(\theta^+)$ no longer depends on w and agrees with CEEX and BabaYaga, in total and separately in the one- and two-photon parts (Section 8). Totals and charge-even distributions are unchanged.

2 Setup

Scenario. KLOE-LA (Strong2020 KLOE-I), $\sqrt{s} = 1.02$ GeV: $50^\circ \leq \theta_\pm \leq 130^\circ$; $|p_\pm^z| > 90$ MeV or $p_\pm^\perp > 160$ MeV; at least one photon with $50^\circ \leq \theta_\gamma \leq 130^\circ$ and $E_\gamma > 20$ MeV; $0.1 \leq M_{\pi\pi}^2 \leq 0.85$ GeV². Angles are measured from the electron beam in both codes. Histograms: the 18 KLOE-I observables with 600 uniform bins. The forward–backward asymmetry is $A = [\sigma(\theta < 90^\circ) - \sigma(\theta > 90^\circ)]/\sigma$.

Codes.

- **CEEX-main** (commit e05d007): `pipig -scenario=KLOE-LA -NLO`, one-loop amplitudes from GoSam (`eepipig_noVP_1L_onshell`, `eepipigg`), YFS soft subtraction with cutoff $E_{\text{min}} = 10^{-5}$ GeV (10^{-4} GeV only in the cutoff checks); 5 MPI ranks per task.
- **PHOKHARA 10.0** (`PHOKARA_10.0 @ 133d68b`) with the KLOE-I cut file, card `nlo=nlo2=1`, `IFSNLO=1`, `ISR+INT+FSR`, `ivac=0`, `FF_pion=-1` ($F_\pi = 1$), no GVMD, no f_0 , soft cutoff $w = 10^{-5}$ (soft region $E_\gamma < w\sqrt{s}$). The production binary `phokhara_nostop` only disables the abort on “Max(i) too small” (Section 10); only weighted events are used.
- **BabaYaga** and the earlier PHOKHARA reference set (5 Aug) are taken from the Strong2020 comparison files.

Pipeline. SLURM arrays per run name; CEEX rank outputs are screened with `check_total.sh` and outliers are moved aside (`quarantine.sh`) before merging; `plot_compare.py` produces the distribution and ratio plots. Everything is in the repository (Appendix A).

3 How PHOKHARA computes $\pi^+\pi^-\gamma$ at NLO

This section describes the pion branch of PHOKHARA 10.0 in the configuration of our card (`pion=1`, `nlo=nlo2=1`, `fsr=2`, `fsrnlo=1`, $F_\pi = 1$, no vacuum polarisation): first the overall structure, then the routines, the UV and IR treatment, and finally the two blocks where the errors are (Section 3.5). Line numbers refer to the unpatched `phokhara_10.0.f` unless another file is named. Statements marked (*inferred*) are our reading of the code; the code itself does not document them.

3.1 Overall structure

Amplitudes. At tree level the one-photon amplitude is $M = I + F$:

- I (ISR): the photon is emitted from the $e^\pm$ line, and $\gamma^*(q^2) \rightarrow \pi\pi$ with the current $J^\mu = (q_+ - q_-)^\mu F_\pi(q^2)$;
- F (FSR): $\gamma^*(s) \rightarrow \pi\pi\gamma$ in scalar QED, i.e. emission from π^+ , from π^- and the seagull (point-like pions).

$|I|^2$ and $|F|^2$ are charge-even; the interference $2\text{Re}[I^*F]$ is charge-odd (odd under $\pi^+ \leftrightarrow \pi^-$) and is the only source of the pion forward–backward asymmetry. At NLO every one-loop and soft term is either charge-even or charge-odd; Table 1 lists them all.

Event classes. Two parameters split the photon phase space:

- w , the soft-photon cutoff: a photon with $E_\gamma < w\sqrt{s}$ (CM frame) is soft, is never generated, and is integrated analytically;
- $\mathbf{gmin}$, the minimum energy of the tagged photon.

Each event is a one-photon or a two-photon event; one random number picks the class with probability proportional to its maximum weight from the scan pass.

- **One-photon events** (one hard photon above $\mathbf{gmin}$) carry in a single weight the tree $|M|^2$, all one-loop virtual corrections, and the integrated soft emission.
- **Two-photon events** carry the exact tree $|M|^2$ for $e^+e^- \rightarrow \pi^+\pi^-\gamma\gamma$: one photon is tagged ($E \geq \mathbf{gmin}$, inside the angular cuts), the other only has to be above $w\sqrt{s}$.

The two classes add up to the full $\mathcal{O}(\alpha)$ correction, and the w dependence must cancel in the sum. This is phase-space slicing; there are no local subtraction terms.

Precision. The one-photon kinematics are generated in quad precision (`momenta_quad.f`). Only `virt_pions` uses the quad momenta; all other pieces use rounded double copies.

3.2 The one-photon weight

Notation. Write the one-photon amplitude to $\mathcal{O}(e^5)$ as

$$M = \underbrace{I + F}_{\text{tree}} + \underbrace{\delta I_e + \delta I_\pi + \delta F_e + \delta F_\pi + \delta B_e + \delta B_\pi}_{\text{one loop}},$$

where the one-loop amplitudes are

- δI_e : ISR with the loop on the electron line (vertex and self-energy corrections around the emitted photon);
- δI_π : ISR with the $\gamma^*\pi\pi$ vertex correction at q^2 ;
- δF_e : FSR with the $e^+e^-\gamma^*$ vertex correction at s ;
- δF_π : FSR with the loop on the pion line (self-energies, vertices, seagull);
- $\delta B_e, \delta B_\pi$: two-photon exchange between the electron and pion lines (boxes and pentagons), with the real photon on the electron or on the pion line.

Figure 1 shows one example diagram for each class.

The two vertex corrections factorise: $\delta F_e = \frac{1}{2}v_s F$ and $\delta I_\pi = \frac{1}{2}v_f I$, with v_s (`ver_s`) and v_f (`ver_f=vertL0`) real $|M|^2$ -level factors.

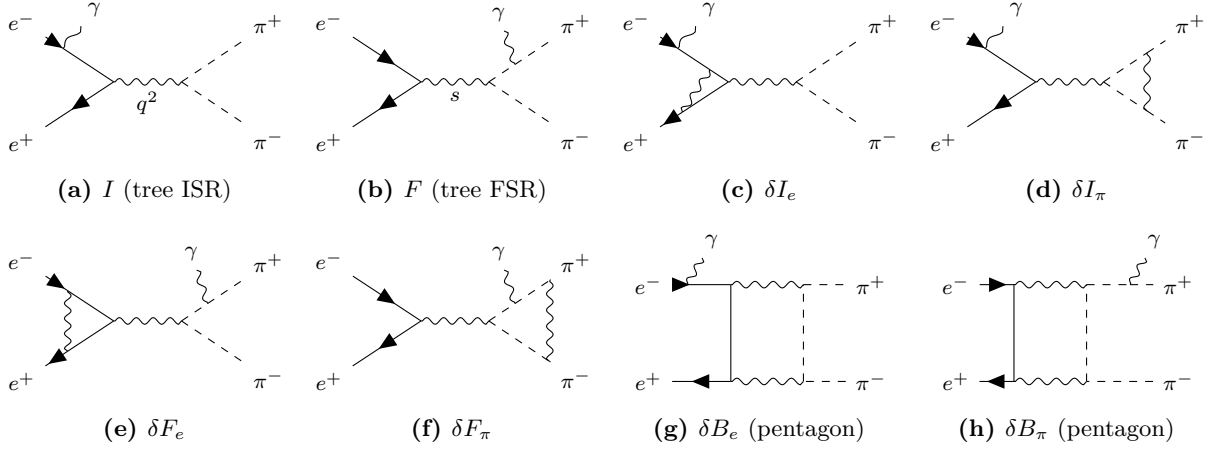


Figure 1: One example diagram per amplitude class of Section 3.2, with the real photon γ ($E_\gamma > w\sqrt{s}$). Top: the tree amplitudes I and F and the ISR corrections; bottom: the FSR corrections and the two-photon exchange. Each class also contains the diagrams with the photon on the other leg, the self-energies, the crossings ($\pi^+ \leftrightarrow \pi^-$, for δB the crossed photon lines) and, for F and δF_π , the seagull $\gamma\gamma\pi\pi$ vertex.

All contributions. Expanding $|M|^2$ and adding the soft photons gives the terms of Table 1. Counting charges ($I \propto Q_e^2 Q_\pi$, $F \propto Q_e Q_\pi^2$, each loop adds two charges) tells which are charge-odd: a term is charge-odd when it has an odd power of Q_π . The tree $|M|^2$ is split over three places: $|I|^2$ in `amplit`, $|F|^2$ in B1, $2\text{Re}[I^*F]$ in B3.

amplit: ISR squared (lines 2069, 6757ff). `amplit` = $L^{\mu\nu} H_{\mu\nu}$ is a contraction of a leptonic and a hadronic tensor, not a helicity sum:

- $H^{\mu\nu} \propto |F_\pi(q^2)|^2 (q_+ - q_-)^\mu (q_+ - q_-)^\nu$ (`HadronicTensorISR`): the $\gamma^* \rightarrow \pi\pi$ vertex, squared;
- $L^{\mu\nu}$ (`LeptonicTensor1ISR`): the electron line with the emitted photon, squared and summed over the photon polarisation. Its coefficients come from `Bornvirtualsoft` (G. Rodrigo et al., EPJC 22 (2001) 81; J.H. Kühn, G. Rodrigo, EPJC 25 (2002) 215): LO plus the $\mathcal{O}(\alpha)$ ISR virtual+soft correction with m_e kept, a renormalised analytic result.

With `nlo2=1` the analytic soft function is then removed (lines 6838–6856), so

$$\text{amplit} = |I|^2 + 2\text{Re}[I^* \delta I_e]_{\text{V+S}} - S_{\text{ISR}}(w) |I|^2 = |I|^2 + 2\text{Re}[I^* \delta I_e]_{\text{V}},$$

where the subscript V means the virtual correction with the IR pole dropped and its soft counterpart supplied by `NLOptions`. `amplit` contains nothing else: no FSR, no interference, and not the pion-vertex correction (that is B2).

helicityampL0 $\rightarrow$ ampL0: everything else at tree level and the factorising corrections (lines 9687–10021).

$$\text{helicityampL0} = \frac{(4\pi\alpha)^3}{4} \text{ampL0} \times (\text{phase space}), \quad \text{ampL0} = \sum_{\lambda_\gamma=1,2} (B_1 + B_2 + B_3 + B_4 + B_5),$$

the $1/4$ being the $e^\pm$ spin average. The amplitudes are 2×2 Weyl blocks A and B, `plus(v) =  $v^0 - \vec{v} \cdot \vec{\sigma}$` and `minus(v) =  $v^0 + \vec{v} \cdot \vec{\sigma}$` :

Amplitude	A block	B block
I (tree ISR, from <code>blocksL0</code>)	<code>ma</code>	<code>mb</code>
F (tree FSR, from <code>dvec</code>)	<code>ddmi</code>	<code>ddpl</code>
I_{1L} (ISR at one loop, from <code>NLO_ISR_no_mass_terms</code>)	<code>ami</code>	<code>apl</code>

Table 1: Every contribution to the one-photon weight and where PHOKHARA computes it (full-NLO pion mode). B1–B5 are the blocks of **ampL0** (Table 2). $S_{\text{ISR}}(w)$ is the analytic ISR soft function of Section 3.4. The hard two-photon emission, the other half of the real correction, is computed separately in **gen_2ph**.

Contribution	Charges	C	Computed in	Remark
<i>Tree</i>				
$ I ^2$	$Q_e^4 Q_\pi^2$	+	amplit	tensor $L^{\mu\nu} H_{\mu\nu}$
$ F ^2$	$Q_e^2 Q_\pi^4$	+	ampL0 B1	
$2\text{Re}[I^* F]$	$Q_e^3 Q_\pi^3$	−	ampL0 B3	only with fsr=2
<i>One loop</i>				
$2\text{Re}[I^* \delta I_e]$	$Q_e^6 Q_\pi^2$	+	amplit	Rodrigo, m_e kept; $-S_{\text{ISR}} I ^2$
$2\text{Re}[I^* \delta I_\pi] = v_f I ^2$	$Q_e^4 Q_\pi^4$	+	ampL0 B2	
$2\text{Re}[F^* \delta F_e] = v_s F ^2$	$Q_e^4 Q_\pi^4$	+	ampL0 B1	
$2\text{Re}[F^* \delta F_\pi]$	$Q_e^2 Q_\pi^6$	+	virt_pions	virtual_FSRHelSum
$2\text{Re}[I^* \delta F_e + F^* \delta I_\pi]$ $= \text{Re}[I^* F](v_s + v_f)$	$Q_e^5 Q_\pi^3, Q_e^3 Q_\pi^5$	−	ampL0 B4	H1
$2\text{Re}[F^* \delta I_e]$	$Q_e^5 Q_\pi^3$	−	ampL0 B5	massless e ; H2
$2\text{Re}[I^* \delta F_\pi]$	$Q_e^3 Q_\pi^5$	−	virt_pions	virtual_FSRHelSum
$2\text{Re}[I^* \delta B_e]$	$Q_e^5 Q_\pi^3$	−	virt_pions	Pentabox_full_isr, cross_ISR
$2\text{Re}[F^* \delta B_e]$	$Q_e^4 Q_\pi^4$	+	virt_pions	same routines
$2\text{Re}[I^* \delta B_\pi]$	$Q_e^4 Q_\pi^4$	+	virt_pions	Pentabox_PIFSR, cross_FSR
$2\text{Re}[F^* \delta B_\pi]$	$Q_e^3 Q_\pi^5$	−	virt_pions	same routines
<i>Soft emission, $E_\gamma < w\sqrt{s}$</i>				
$ M_{\text{tree}} ^2 \times (ee, \pi\pi \text{ eikonal})$	$\times Q_e^2, \times Q_\pi^2$	$\pm$	NLOptions	odd part from $2\text{Re}[I^* F]$
$ M_{\text{tree}} ^2 \times (e\pi \text{ eikonal})$	$\times Q_e Q_\pi$	$\mp$	NLOptions	odd part from $ I ^2 + F ^2$

The beam spinors enter through four fixed combinations of the blocks (lines 9939–9942), with $e_b = E + |\vec{p}|$ of the positron:

$$\begin{aligned}
T_1(X) &= m_e(B_{11} - A_{11}), & T_3(X) &= -e_b A_{21} + \frac{m_e^2}{e_b} B_{21}, \\
T_2(X) &= m_e(B_{22} - A_{22}), & T_4(X) &= e_b B_{12} - \frac{m_e^2}{e_b} A_{12}.
\end{aligned}$$

(*Inferred*) T_1, T_2 are the m_e -suppressed helicity-flip amplitudes and T_3, T_4 the two leading helicity-conserving ones. With $\langle X, Y \rangle \equiv \sum_{k=1}^4 T_k(X)^* T_k(Y)$ and $|X|^2 \equiv \langle X, X \rangle$, the five blocks are listed in Table 2.

Table 2: The blocks of **ampL0** (pion branch), per photon polarisation. Vacuum-polarisation factors **cvac** are 1 in our setup and omitted.

Block	Code computes	Physics content	Condition	Lines
B1	$ F ^2(1 + v_s)$	$ F ^2 + 2\text{Re}[F^* \delta F_e]$	always ^a	9938–9943
B2	$ I ^2 v_f$	$2\text{Re}[I^* \delta I_\pi]$	fsrnlo=1	9945–9951
B3	$2\text{Re}\langle I, F \rangle$	$2\text{Re}[I^* F]$	fsr=2	9954–9966
B4	$\text{Re} \sum_{k \leq 3} T_k(I)^* T_k(F)$ $+ \text{Re}[T_4(I)^* T_4(F)](v_f + v_s)$	intended $\text{Re}\langle I, F \rangle(v_f + v_s)$	nlo2=1, pion=1	9969–9982 H1
B5	$\text{Re}\langle F, I_{1L} \rangle$, with $I_{1L} = I(1 + S_{\text{ISR}}) + 2\delta I_e$	intended $2\text{Re}[F^* \delta I_e]$	nlo2=1, pion=1	9984–10007 H2

^a $v_s = 0$ if **fsrnlo=0**.

So **helicityampL0** holds the tree $|F|^2$ and $2\text{Re}[I^* F]$ and all one-loop terms that are built from tree amplitudes: the two factorising vertex corrections (B1, B2, B4) and the ISR loop interfered with the FSR tree (B5). As coded, B4 and B5 carry two extra terms, both proportional to the tree interference:

$$\Delta B_4 = \text{Re} \sum_{k \leq 3} T_k(I)^* T_k(F) (1 - (v_f + v_s)), \quad \Delta B_5 = \text{Re}\langle I, F \rangle (1 + S_{\text{ISR}}(w)).$$

The decomposition $I_{1L} = I(1 + S_{\text{ISR}}) + 2\delta I_e$ is read from `V1(1)` (Section 7.2) and confirmed numerically by the per-point test (Section 6.4).

NLOptions: soft emission (`hard_soft_part/one_photon.f`).

$$|M_{\text{tree}}|^2 \times 2[S_{jj}^{ee} - S_{12}^{ee} + S_{jj}^{\pi\pi} - S_{12}^{\pi\pi} + (-S_{11}^{e\pi} - S_{22}^{e\pi} + S_{12}^{e\pi} + S_{21}^{e\pi})],$$

with the full tree $|M_{\text{tree}}|^2 = |I|^2 + |F|^2 + 2\text{Re}\langle I, F \rangle$ recomputed in helicity form, and the eikonal integrals of `soft_integrals.f` ('t Hooft–Veltman, eq. (7.3)) evaluated on the event momenta: `pjpj`, `p1p2` (ee), `qjqj_sub`, `q1q2_sub` ($\pi\pi$) and the four $e\pi$ interference terms.

virt_pions: the loops that do not factorise (`virtualpions.f`, quad precision, $\mu^2 = s$). Each loop amplitude is contracted with a helicity vector of the full tree $I + F$ (the `hel()` arrays carry both the $F_\pi(s)$ and the $F_\pi(q^2)$ parts), so each routine gives both the charge-even and the charge-odd rows of Table 1. The diagram content is inferred from the propagator masses:

- `Pentabox_full_isrHelSum`, `cross_ISRHelSum`: δB_e , masses $(m_e, m_e, 0, m_\pi, 0)$, and its $\pi^+ \leftrightarrow \pi^-$ crossing;
- `Pentabox_PIFSRHelSum`, `cross_FSRHelSum`: δB_π , masses $(m_e, 0, m_\pi, m_\pi, 0)$, and its crossing;
- `virtual_FSRHelSum` and `...cr (test_vFSR.f)`: δF_π , only massless and m_π propagators, with QCDLoop integrals.

3.3 UV renormalisation

- **Analytic pieces.** `Bornvirtualsoft`, v_s and v_f (`vertL0`) are closed-form, on-shell renormalised results: electron vertex at s and point-like pion form factor at q^2 , normalised to the charge e at $q^2 = 0$. No counterterm appears in the code; with vacuum polarisation off, α does not run.
- **Pion-line loops** (`test_vFSR.f:740–761`). This is the only place with explicit counterterms: on-shell scalar-QED counterterms built from B_0 , C_0 and the derivative $B'_0(m_\pi^2)$ (wave-function renormalisation), multiplied by tree helicity structures (`diag1`, `diag2`) and added to the result (`counter1`, `counter3`, `counter8`, `counter95`). The crossed version has no `diag1` analogue.
- **Boxes and pentagons.** The two-photon-exchange set is UV-finite and has no counterterms. Tensor integrals are reduced with Passarino–Veltman in quad precision; scalar integrals come from QCDLoop.
- **Regularisation.** Dimensional regularisation with $\mu^2 = s$; the $1/\epsilon$ poles are dropped and the finite parts kept. The IR poles cancel against the soft integrals, written in the same convention with $\lambda = \sqrt{s}$. Scale shifts give $\ln(q^2/s)$ terms, e.g. the extra $-\frac{1}{2}\ln(q^2/s)$ in `qjqj_sub` and `q1q2_sub`.

3.4 IR treatment

Slicing. Soft region $E_\gamma < w\sqrt{s}$ (CM frame): eikonal approximation, integrated analytically in `NLOptions`. Hard region: exact two-photon matrix element (`gen_2ph`), with the ISR·ISR (`ma/mb`), ISR·FSR (`mapi`) and FSR·FSR (`Jplus/Jmin`) amplitudes added coherently and squared with the same T_1 – T_4 pattern.

The full-NLO rewrite (`nlo2=1`). In the older NLO mode each analytic vertex result was “virtual + soft”, with its own analytic $\ln(2w)$ soft function (ISR in the leptonic tensor, v_s for the electron vertex, `vertL0` for the pion vertex). That scheme has no e – π soft interference. `nlo2=1` replaces all soft emission by the exact eikonal integrals of `NLOptions`, which include the $e\pi$ terms, so every analytic soft function has to be removed from its “V+S” formula. The code does this in three places:

- leptonic tensor (lines 6838–6856): $L - L_{\text{LO}} S_{\text{ISR}}(w)$, with

$$S_{\text{ISR}}(w) = \frac{\alpha}{\pi} \left[-\ln(4w^2) (1 + L) - \frac{1}{2}L^2 - L - \frac{\pi^2}{3} \right], \quad L = \ln \frac{m_e^2}{s};$$

- v_s (lines 2754–2761): $v_s - S_{\text{ISR}}(w)$; the $\ln(2w)$ terms cancel exactly;
- `vertL0` (line 9795): virtual part minus the soft function of Nucl. Phys. B57 (1973) 381, eq. (29), “without terms proportional to $\text{Log}(2w)$ ”.

What must hold. After the rewrite every one-loop piece should be free of analytic w terms. All w dependence then comes from `NLOptions` and cancels against the two-photon events. A leftover analytic $\ln w$ term does not cancel, and the result depends on w .

3.5 Where the errors are

Both errors are in `ampL0`, in the two blocks of Table 2 that correct the tree interference, and both add a multiple of the tree interference (the $\Delta B_4, \Delta B_5$ of Section 3.2):

- **B4 (H1).** The vertex factor ($v_f + v_s$) is placed inside `dreal()` after the last helicity term, so it multiplies only T_4 . T_1 – T_3 of the tree interference are added a second time with coefficient 1. T_3 is a leading helicity amplitude, so this is an $\mathcal{O}(1)$ extra charge-odd term, not an $\mathcal{O}(\alpha)$ one.
- **B5 (H2).** `NLO_ISR_no_mass_terms` returns the ISR amplitude *including* the tree, not the loop alone: its leading coefficient `V1(1)` contains the tree “1” and the analytic soft function $S_{\text{ISR}}(w)$. The “1” repeats half of B3, and $S_{\text{ISR}}(w)$ is the one soft function the full-NLO rewrite never removed, which breaks the condition of Section 3.4.

Both extra terms are proportional to $\text{Re}\langle I, F \rangle$, which is charge-odd. They integrate to zero over the charge-symmetric KLOE-LA cuts, so σ and all charge-even distributions are unaffected; only the pion polar angles move. Section 7 gives the code and the fixes.

Oddities without effect here. `soft_f` in `vertL0` and `ver_f` in `ampNLO` are computed but not used. The virtual-FSR term in `virt_pions` has an explicit prefactor four times that of the pentagons ($32\pi^2\alpha^4$ vs $8\pi^2\alpha^4$); we did not check whether this is compensated inside the routine, but the per-point test (Section 6.4) agrees with CEEX to 2.5×10^{-5} of the tree. Some “(-1)” comments do not match the code.

4 The discrepancy

Table 3: Before the fix. PHOKHARA (unpatched) 300 seeds $\times$ 500k events; CEEX more1M, 148 tasks $\times$ 5 ranks $\times$ 1M events at $E_{\text{min}} = 10^{-5}$ GeV (732 rank outputs after discarding 8 whose total diverged).

	σ [nb]	$A(\theta^+)$
CEEX, $E_{\text{min}} = 10^{-5}$ GeV	0.268389(9)	0.23725(3)
BabaYaga	0.268389(4)	0.23724(1)
PHOKHARA, this work	0.268353(46)	0.27872(16)
PHOKHARA, earlier reference	0.268393(23)	0.27861(8)

The new PHOKHARA run reproduces the earlier reference. All generators pass the charge-conjugation mirror test $\theta^+(\theta) = \theta^-(180^\circ - \theta)$, so this is not a mislabelled charge. Figure 2 shows the θ^+ distribution and Figure 3 the ratio of every observable to PHOKHARA: only the three pion angles are off, by a smooth tilt from 0.95 to 1.28.

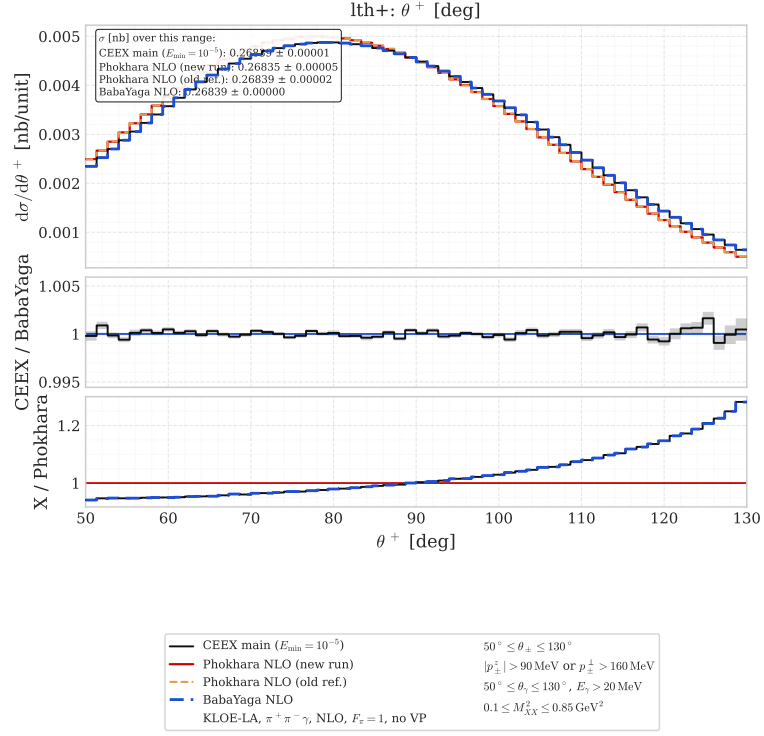


Figure 2: Before the fix: unpatched PHOKHARA (300×500k) against CEEEX more1M (148×5×1M): $d\sigma/d\theta^+$, CEEEX/BabaYaga, and CEEEX and BabaYaga over PHOKHARA (60 bins). CEEEX ($E_{\min} = 10^{-5}$ GeV, black) and BabaYaga (blue) agree; PHOKHARA (red) is tilted.

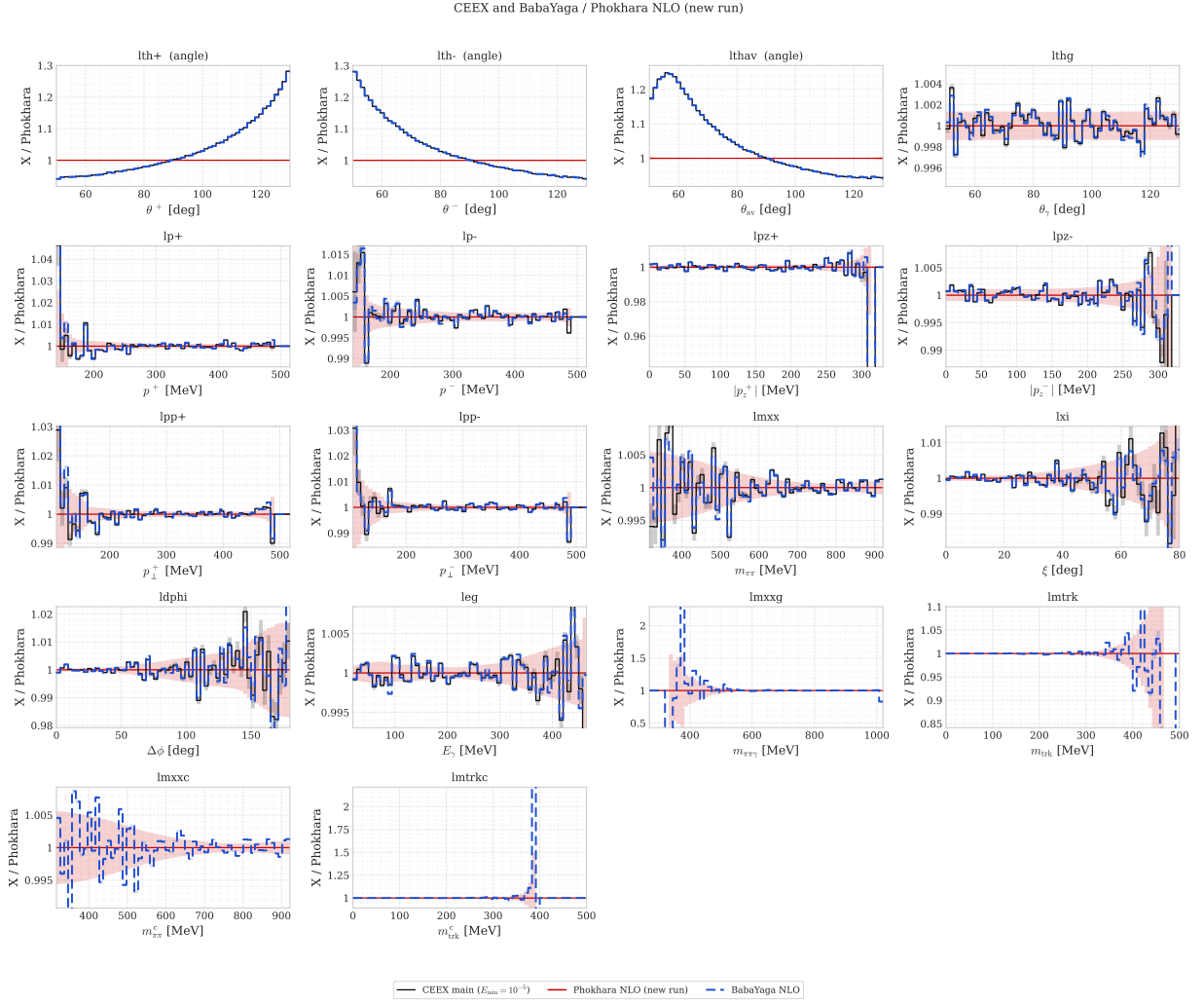


Figure 3: Before the fix: every observable, CEEX and BabaYaga over PHOKHARA. Only θ^+ , θ^- and θ_{avg} are off; the spikes in $M_{\pi\pi\gamma}$ and M_{trk} are empty tail bins.

At LO the codes agree exactly (earlier runs with the Strong2020 form factor: $A_{\text{LO}} = 0.0779$ and $\sigma_{\text{LO}} = 2.6523$ nb in both), so the difference lies in the NLO correction to the charge-odd part of the cross section.

5 Strategy

The difference is charge-odd (C -odd): it flips sign under $\pi^+ \leftrightarrow \pi^-$ and integrates to zero over the symmetric cuts. At $\mathcal{O}(\alpha^4)$ the C -odd part of the cross section consists of the gauge-invariant classes $Q_e^5 Q_\pi^3$ and $Q_e^3 Q_\pi^5$ (the tree interference is $Q_e^3 Q_\pi^3$). Three tools were used to isolate it:

- **Cutoff independence.** The split into one-photon (Born + virtual + soft photons below the cutoff) and two-photon (hard) parts depends on the cutoff; the sum must not.
- **Matched split.** For the same cutoff each part is uniquely defined, so the two codes must agree part by part. PHOKHARA $w = 9.804 \times 10^{-6}$ ($w\sqrt{s} = 10^{-5}$ GeV) matches CEEX $E_{\min} = 10^{-5}$ GeV.
- **Momentum swap.** At a phase-space point, the C -odd part of any piece X is $X_{\text{odd}} = \frac{1}{2}[X(p_+, p_-) - X(p_-, p_+)]$. No charge parameter is needed.

Each histogram-level test used enough statistics to resolve $A(\theta^+)$ to $\sim 10^{-3}$ while checking that σ and $M_{\pi\pi}$ are unchanged. The plan with all tests and hypotheses is in `plan.txt`.

6 Tests

6.1 T1: soft-cutoff scan

Table 4: Unpatched PHOKHARA, 30 seeds $\times$ 100k per value of w .

w	10^{-3}	10^{-4}	10^{-5}	10^{-6}
$\sigma_{1\gamma}$ [nb]	0.1386	0.0955	0.0530	0.0103
$\sigma_{2\gamma}$ [nb]	0.1297	0.1728	0.2149	0.2580
σ [nb]	0.26833(33)	0.26827(35)	0.26792(33)	0.26830(30)
$A(\theta^+)$	0.3106(11)	0.2950(12)	0.2785(12)	0.2637(11)

$A(\theta^+)$ falls linearly in $\ln w$, by 0.0156 per decade, while σ is constant to 0.1%: the charge-even soft logarithms cancel between the one- and two-photon parts, a charge-odd one does not. In CEEX, $E_{\min} = 10^{-4}$ and 10^{-5} GeV give the same $A(\theta^+)$ to 10^{-4} .

6.2 T2: diagnostic variants

Three diagnostic PHOKHARA builds were run at $w = 10^{-4}, 10^{-5}, 10^{-6}$ (Figure 4a):

- **H1 fixed** (the parenthesis of Section 7.1): $A(\theta^+)$ moves by -0.037 and keeps its slope. H1 is a w -independent error.
- **B5 soft log removed** (probe): $A(\theta^+)$ becomes flat in w , but at 0.355. The whole w -dependence comes from this term; removing only the logarithm is not the right correction.
- **Both:** flat, at 0.319.

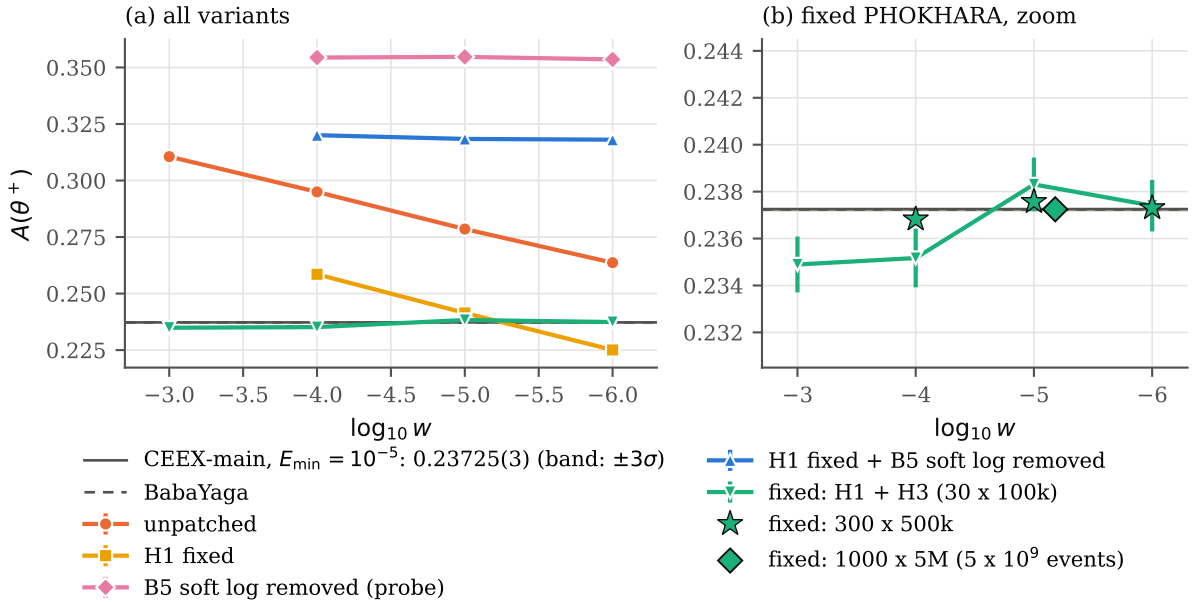


Figure 4: $A(\theta^+)$ against the PHOKHARA soft cutoff for the unpatched code, the T2 diagnostic variants and the fixed code (H1 + H3), with CEEX (grey band: $\pm 3\sigma$) and BabaYaga. The zoom (b) shows the fixed code at $30 \times 100k$, at $300 \times 500k$ (stars) and at $1000 \times 5M$ (diamond, drawn just left of $w = 10^{-5}$).

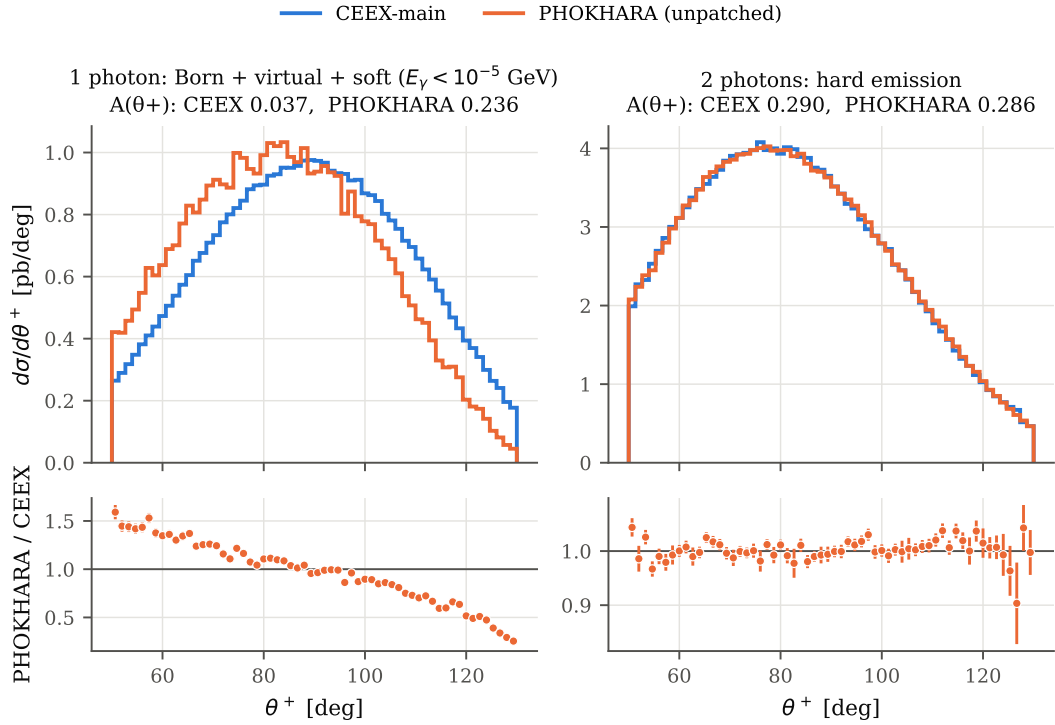
6.3 T3: split by photon multiplicity

CEEX-main was given an opt-in output of the per-multiplicity histograms (CEEX_NPH_HISTOS=1); PHOKHARA writes them with `print_summed=1`.

Table 5: Matched cutoff (10^{-5} GeV). CEEX 10 seeds $\times$ 5 ranks $\times$ 100k; PHOKHARA 30 $\times$ 100k (unpatched).

	σ [nb]		$A(\theta^+)$		angle shapes
	CEEX	PHOKHARA	CEEX	PHOKHARA	χ^2/ndf
1 photon (Born+V+S)	0.05268(7)	0.05239(24)	0.0375(13)	0.2360(44)	79–94
2 photons (hard)	0.21560(37)	0.21588(23)	0.2901(16)	0.2855(10)	1.0–1.4

The one-photon parts have the same σ and the same $M_{\pi\pi}$, θ_γ and E_γ shapes, but charge-odd cross sections of 0.0020 (CEEX) and 0.0124 nb (PHOKHARA). The difference, 0.0104 nb, is the whole total discrepancy (Figure 5). The hard two-photon parts agree.

**Figure 5:** T3: $d\sigma/d\theta^+$ of the one-photon (left) and two-photon (right) contributions at a matched cutoff, with the ratio PHOKHARA/CEEX.

6.4 T4: point-by-point comparison

988 accepted one-photon KLOE-LA points were dumped by a PHOKHARA variant (`phokhara_dump`) together with every piece of the one-photon matrix element, at the event and with the pion momenta swapped. A CEEX-main driver (`t4_ceex`) evaluates the tree, the hard virtual $\frac{\alpha}{4\pi}D$ (GoSam finite part plus mass counterterm), the wave-function counterterm and the YFS factor at the same momenta; the pion mass is set to PHOKHARA’s value. Each code is normalised to its own charge-even tree:

- tree C -odd/ C -even ratio: identical to 7×10^{-13} ;
- C -even NLO correction: CEEX -0.77566 , PHOKHARA -0.77570 , per-point difference rms 10^{-5} ;
- C -odd NLO correction: per-point difference D with rms 0.099;
- C -odd soft (eikonal/YFS) part: agrees, rms 7×10^{-4} .

D is exactly proportional to the tree interference at every point. Subtracting the two errors, each computed exactly per point ($v_s = 0.29655$, $v_f = 0.023\text{--}0.036$, $S_{\text{ISR}}(w) = -0.95556$),

$$\text{pred}_{\text{H1}} = B_4 - \frac{1}{2}B_3(v_f + v_s), \quad \text{pred}_{\text{H2}} = \frac{1}{2}B_3(1 + S_{\text{ISR}}(w)),$$

leaves a residual of rms 2.5×10^{-5} with mean 10^{-6} (Figure 6). H1 alone leaves 1.2×10^{-2} , H2 alone 8.8×10^{-2} .

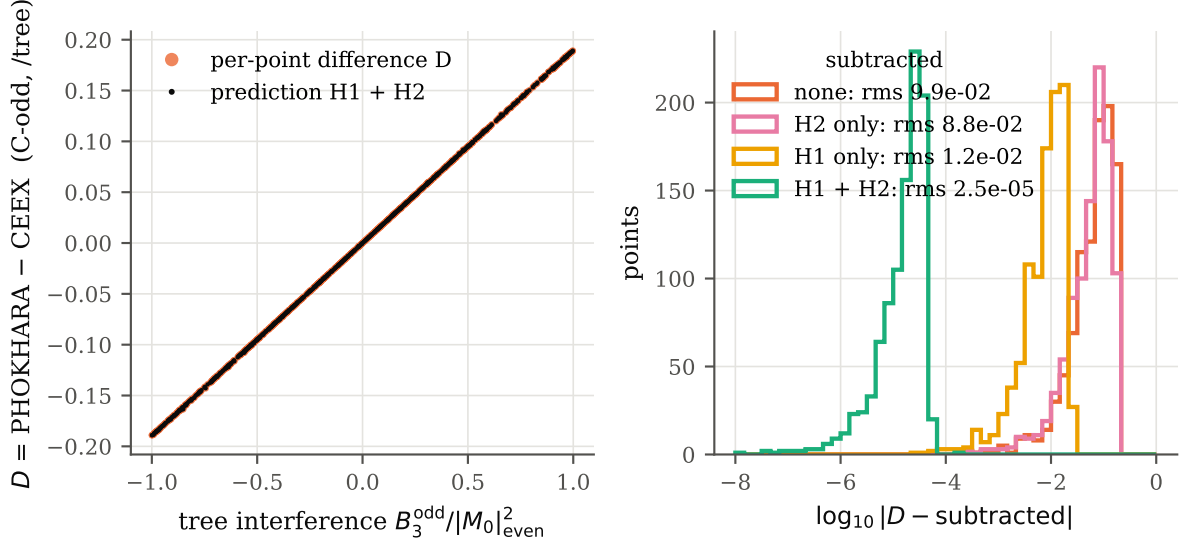


Figure 6: T4, 988 points. Left: the C -odd difference D against the tree interference, with the H1+H2 prediction. Right: $\log_{10} |D - \text{subtracted}|$ for no subtraction, H1, H2 and H1+H2.

7 The two errors in PHOKHARA

The structure of `ampL0` and the blocks B1–B5 are described in Section 3.2 and Table 2. Here I and F are the ISR and FSR tree amplitudes, v_s the electron-vertex correction at s and v_f the pion-vertex correction at q^2 .

7.1 H1: misplaced factor in the $\text{ISR} \times \text{FSR}$ correction

```

if((nlo2.eq.1).and.(pion.eq.1))then
  ampL0 = ampL0
  4    + dreal(
  5      dconjg(dme*(mb(1,1)-ma(1,1))*cvac_qq)*dme*(ddpl(1,1)-ddmi(1,1))*cvac_s    ! T1
  6      + dconjg(dme*(mb(2,2)-ma(2,2))*cvac_qq)*dme*(ddpl(2,2)-ddmi(2,2))*cvac_s    ! T2
  7      + dconjg((-ebppb*ma(2,1)+e1_m2/ebppb*mb(2,1))*cvac_qq)
  8      *(-ebppb*ddmi(2,1)+e1_m2/ebppb*ddpl(2,1))*cvac_s                        ! T3
  9      + dconjg((ebppb*mb(1,2)-e1_m2/ebppb*ma(1,2))*cvac_qq)
  1     *(ebppb*ddpl(1,2)-e1_m2/ebppb*ddmi(1,2))*cvac_s                        ! T4
  2     *(ver_f+ver_s) )                ! multiplies T4 only

```

(Continuation lines re-indented for reading; the code is unchanged.) The block is meant to be $\text{Re}[T_1 + T_2 + T_3 + T_4](v_f + v_s)$, the amplitude-level corrections on each side of $2\text{Re}[I^*F]$. As written it is $\text{Re}[T_1 + T_2 + T_3] + \text{Re}[T_4](v_f + v_s)$, so $\text{Re}[T_1 + T_2 + T_3]$ (about a third of the tree interference) is added with coefficient 1. The muon version of the same block (`phokhara_10.0.f:10384-10394`) is correct: `2*dreal(sum)*(ver_s+ver_f)/2`.

Fix (variants/h1.patch): `*cvac_s *(ver_f+ver_s) )` $\rightarrow$ `*cvac_s )*(ver_f+ver_s)`.

7.2 H2: tree and analytic ISR soft function in the ISR one-loop block

Block B5 contracts the FSR tree with the ISR current at one loop:

```
ampL0=ampL0+dreal( dconjg(... ddpl, ddmi ...)*( ... apl, ami ... )    ! B5
```

The current (`hard_soft_part/ISRcurrent_no_mass_terms.f`) has the coefficient

```
V1(1)=2.d0*log(4.d0*w**2)*(log(m2)+1.d0) -4.d0*log(qq2)+3.d0*log(m2) + ...    ! 1.134
V1(1)=(1.d0-2.d0*alpha/4.d0/pi*V1(1))/Sp    ! 1.210
```

It therefore contains the tree (the “1”) and the analytic ISR soft function, and B5 carries an extra $\text{Re}[F^*I](1 + S_{\text{ISR}}(w))$ with

$$S_{\text{ISR}}(w) = \frac{\alpha}{\pi} \left[-\ln(4w^2) (1 + L) - \frac{1}{2}L^2 - L - \frac{\pi^2}{3} \right], \quad L = \ln \frac{m_e^2}{s}.$$

- The tree interference is already added by the `fstr=2` block (B3).
- In the full-NLO mode the analytic ISR soft function is replaced by the eikonal in `NLOptions`; it is subtracted from v_s , v_f and the leptonic tensor, but not here.
- At $w \simeq 10^{-5}$ the two parts nearly cancel ($1 + S_{\text{ISR}} = 0.044$), but S_{ISR} changes by 0.152 per decade of w . This gives the T1 slope: $\frac{1}{2} \cdot 0.152 \cdot A_{\text{LO}} \approx 0.019$ per decade, against 0.0156 measured.

Fix (`variants/h3.patch`): subtract $\text{Re}[\sum T](1 + S_{\text{ISR}})$ after B5, with S_{ISR} computed as in the v_s subtraction.

8 Result with the fix

`phokhara_fix` is the production binary with `h1.patch` and `h3.patch`.

Table 6: Fixed PHOKHARA against the soft cutoff.

w	10^{-3}	10^{-4}	10^{-5}	10^{-6}
<i>30 seeds $\times$ 100k</i>				
σ [nb]	0.26799(33)	0.26892(35)	0.26854(32)	0.26784(31)
$A(\theta^+)$	0.2349(12)	0.2352(13)	0.2383(12)	0.2374(11)
<i>300 seeds $\times$ 500k</i>				
σ [nb]	–	0.268345(48)	0.268351(46)	0.268354(44)
$A(\theta^+)$	–	0.23682(17)	0.23757(17)	0.23730(16)
<i>1000 seeds $\times$ 5M (5×10^9 events)</i>				
σ [nb]	–	–	0.268398(8)	–
$A(\theta^+)$	–	–	0.23725(3)	–

At high statistics the slope in w is -0.00023 ± 0.00012 per decade (unpatched: $+0.0156$), and the w -averaged asymmetries are

	$A(\theta^+)$	$-A(\theta^-)$	$-A(\theta_{\text{avg}})$
PHOKHARA, fixed (300 $\times$ 500k, w -average)	0.23724(10)	0.23722(10)	0.26651(9)
PHOKHARA, fixed (1000 $\times$ 5M, $w = 10^{-5}$)	0.23725(3)	0.23728(3)	0.26651(3)
CEEX, $E_{\text{min}} = 10^{-5}$ GeV (more1M, 148 $\times$ 5 $\times$ 1M)	0.23725(3)	0.23726(3)	0.26648(3)
BabaYaga	0.23724(1)	0.23727(1)	0.26650(1)

At $w = 10^{-5}$ and 10^{-6} the fixed PHOKHARA agrees with CEEX and BabaYaga within $0.3\text{--}1.5\sigma$. The $w = 10^{-4}$ sample lies 0.0007 lower. To check it, the multiplicity split of Section 6.3 was repeated with the fixed code at a matched cutoff of 10^{-4} GeV (PHOKHARA $w = 9.804 \times 10^{-5}$, 100 seeds $\times$ 500k; CEEX $E_{\text{min}} = 10^{-4}$ GeV):

	CEEX	σ [nb] PHOKHARA (fixed)	CEEX	$A(\theta^+)$ PHOKHARA (fixed)
1 photon (Born+V+S)	0.09533(10)	0.09539(7)	0.1548(11)	0.1558(7)
2 photons (hard)	0.17290(23)	0.17301(5)	0.2826(13)	0.2829(2)
sum	0.26823(25)	0.26840(8)	0.2372(9)	0.2377(3)

Both parts agree, with angle-shape $\chi^2/\text{ndf} = 0.7\text{--}1.4$, and this sample gives $A(\theta^+) = 0.2377(3)$, consistent with $w = 10^{-5}$ and 10^{-6} . A 2% change of the cutoff cannot move $A(\theta^+)$ by 0.0009, so the low $w = 10^{-4}$ value is a fluctuation of that sample (2.6σ from this one). T4 at $w = 10^{-4}$ also leaves no systematic C -odd residual after H1+H2 (mean -7×10^{-6}).

Highest statistics. A final run of 1000 seeds $\times$ 5M events (5×10^9 events passing the cuts, $w = 10^{-5}$) gives $\sigma = 0.268398(8)$ nb and $A(\theta^+) = 0.23725(3)$, $-A(\theta^-) = 0.23728(3)$, $-A(\theta_{\text{avg}}) = 0.26651(3)$. To match this precision on the CEEX side, a further CEEX run (more1M: 5 MPI ranks $\times$ 1M events per task at $E_{\text{min}} = 10^{-5}$ GeV, seeds 101–300, independent of prod500k) was started; the 148 tasks finished at the time of writing (732 rank outputs; 8 whose total diverged were discarded by `check_total.sh`) give $\sigma = 0.268389(9)$ nb and

$$A(\theta^+) = 0.23725(3), \quad -A(\theta^-) = 0.23726(3), \quad -A(\theta_{\text{avg}}) = 0.26648(3),$$

in agreement with the fixed PHOKHARA and with BabaYaga to 3×10^{-5} in all three asymmetries. Figures 7 and 8 repeat Figures 2 and 3 with these two runs: the pion-angle shape χ^2/ndf of CEEX against the fixed PHOKHARA drops from ~ 1900 to 0.8–1.4, and against BabaYaga it is 0.9–1.2.

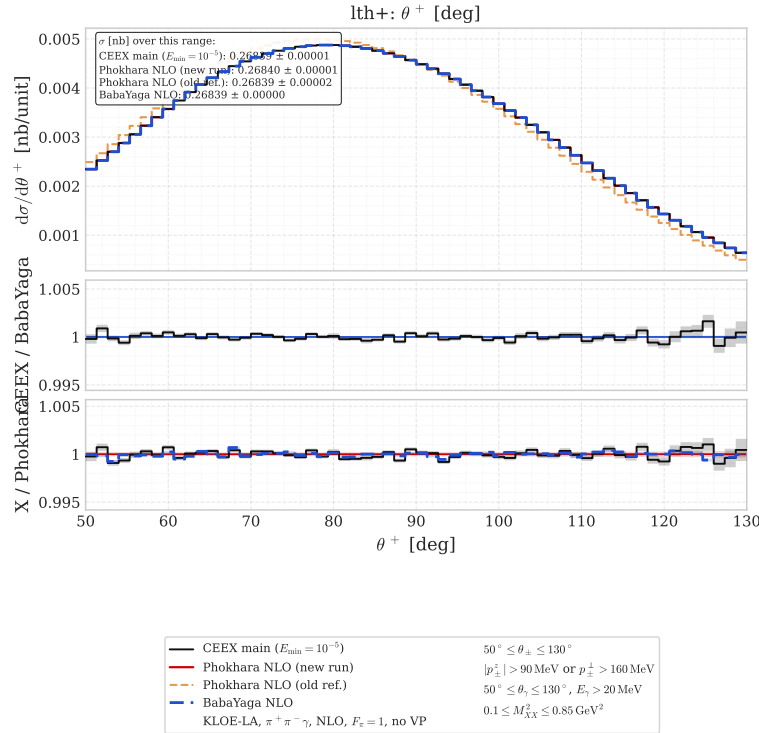


Figure 7: After the fix: as Figure 2, with the fixed PHOKHARA (1000 $\times$ 5M, $w = 10^{-5}$) as the denominator and CEEX more1M (148 $\times$ 5 $\times$ 1M, $E_{\text{min}} = 10^{-5}$ GeV). The tilt is gone.

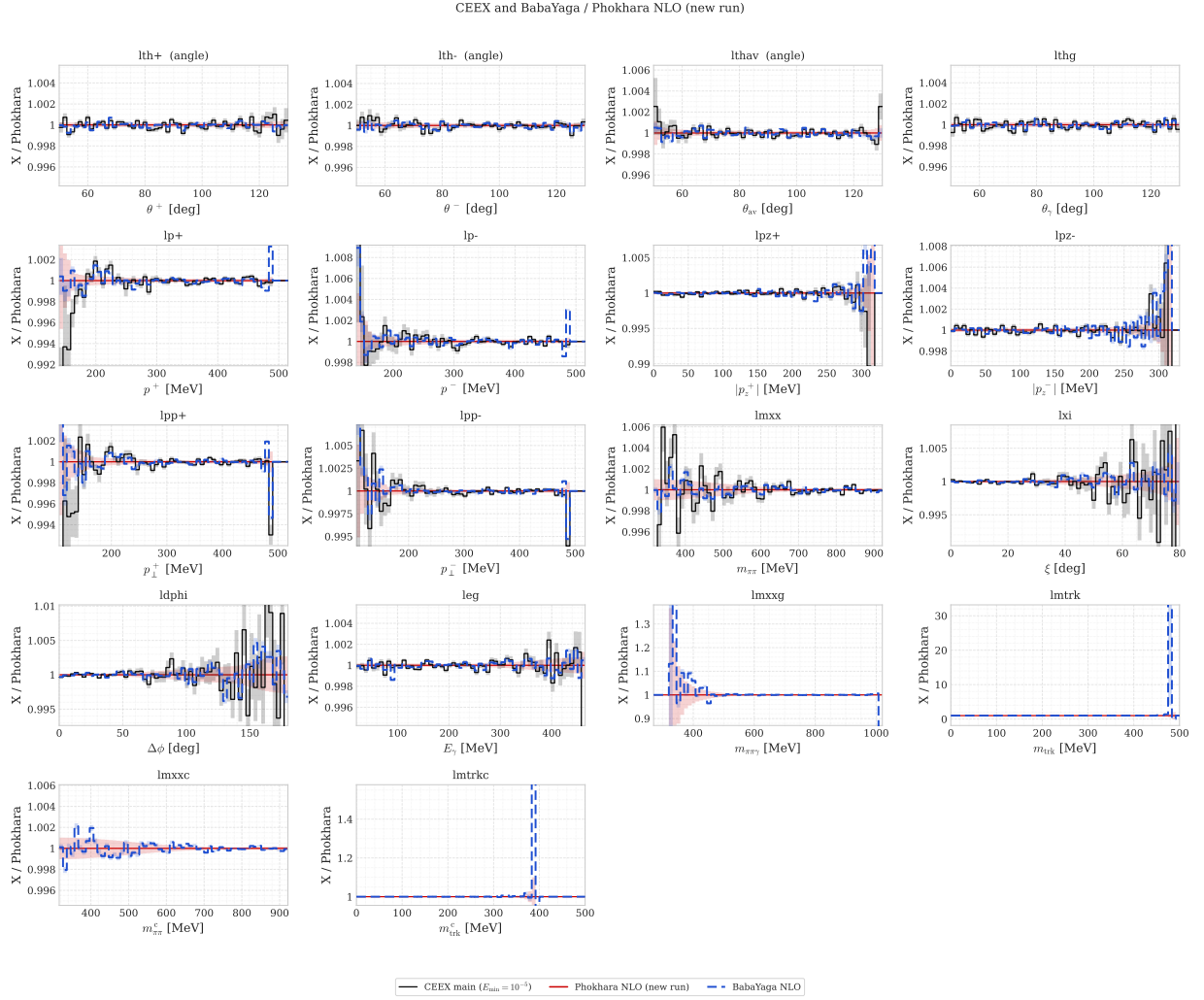


Figure 8: After the fix: every observable, CEEX (more1M, $E_{\min} = 10^{-5}$ GeV) and BabaYaga over the fixed PHOKHARA (1000×5M).

9 Caveats and open points

- Only the configuration studied here was tested ($F_\pi = 1$, no VP, KLOE-LA cuts). Both errors are on code paths active for `pion=1` with `nlo2=1` in any configuration, so they should also affect the Strong2020 form-factor results (there the NLO gap in $A(\theta^+)$ is 0.012: CEEX 0.0804, PHOKHARA 0.0924). This has not been re-run.
- After the fix, one of the three high-statistics samples ($w = 10^{-4}$) lies 0.0007 below the others; an independent sample at the same cutoff agrees with them and with CEEX part by part (Section 8), so it is treated as a fluctuation.
- Muons take a different, correct branch and are not affected.
- Totals and charge-even distributions are not affected by either error (σ unchanged in every run; charge-even per-point agreement 10^{-5}).
- The 0.00027 (2σ) excess of CEEX prod500k over BabaYaga in $A(\theta^+)$ was a statistical fluctuation: CEEX more1M gives $A(\theta^+) = 0.23725(3)$, equal to BabaYaga (0.23724(1)) and the fixed PHOKHARA (0.23725(3)). more1M was still running when this report was written (148 of 200 tasks); the numbers will be updated with the full run.
- With 5×10^9 PHOKHARA events, small BabaYaga–PHOKHARA shape differences become visible in a few charge-even observables (χ^2/ndf 2–3 in $p_\pm$, $p_\pm^\perp$, E_γ ; 10 in $|p_-^z|$, where

BabaYaga’s binning already gave ~ 20 before). CEEX agrees with PHOKHARA there at $\chi^2/\text{ndf} \approx 1$. They are unrelated to the pion-angle issue.

- Tests T5 (charge-class split) and T6 (two-photon point-by-point) of the plan were not needed: the two-photon part agrees (T3) and H1+H2 explain the one-photon part (T4).

10 Other findings

- The checked-out PHOKARA_10.0 has an empty stub `variables_cuts.f` (no cuts, no histograms); the KLOE-I cut file comes from the Strong2020 scenario repository. The card format and the FF_pion numbering changed with respect to the Strong2020 cards.
- PHOKHARA aborts without writing histograms when a weight exceeds the scanned maximum (`phokhara_10.0.f:1910,2135,2263`). Dropping such seeds, as the Strong2020 merge did, biases the result; for weighted events the abort is unnecessary.
- `virt_pions` overwrites its quad-precision momentum arguments (~ 1 GeV). This is harmless in production but breaks any second call in the same event.
- In the 5 Aug reference folder, `Phokhara/NL0novp_*.csv` is the muon result (0.9925 nb), not the pion one.
- CEEX: a few rank outputs per hundred diverge at 100k events (11/300); they are removed before merging. The CEEX-main README still documents `-ref_alloc` and `-vfreeze`, which main no longer parses.
- Cost: at equal precision on σ (prod100k), PHOKHARA needed about 45 times less CPU than CEEX: 0.25 against 28 CPU-hours.

A Reproduction and files

<code>plan.txt</code>	test plan (T0–T6, hypotheses H1–H5)
<code>RUN/config.sh</code>	all paths
<code>RUN/RUN_CEEEX/</code> , <code>RUN/RUN_PHOKHARA/</code>	SLURM submitters and job scripts, runcards
<code>RUN/RUN_PHOKHARA/variants/</code>	<code>h1.patch</code> , <code>h3.patch</code> (the fix), <code>h2.patch</code> (T2 probe), <code>dump.patch</code> (T4), <code>build.sh</code>
<code>RUN/RUN_CEEEX/t4/</code>	CEEX point driver; <code>MC_nphsplit.patch</code> (per-multiplicity histograms)
<code>RUN/POST_PROCESS/</code>	<code>merge</code> , <code>outlier check</code> , <code>phok_table.py</code> , <code>t3_compare.py</code> , <code>t4_compare.py</code>
<code>RESULTS/</code>	<code>prod500k</code> , <code>t1_t2</code> , <code>t3_split</code> , <code>t4_points</code> , <code>t4_fix</code> , <code>fix500k</code> , <code>final500k</code> , <code>fix5M</code> , <code>final5M</code> , <code>final5M_c1M_e5</code> , <code>before_c1M_e5</code>
<code>REPORT/</code>	this report, <code>make_figures.py</code> , <code>make_numbers.py</code>

Building the fixed binary:

```
cd RUN/RUN_PHOKHARA/variants
./build.sh fix h1.patch h3.patch          # -> src/phokhara_fix
cd ../..
EXE=RUN_PHOKHARA/src/phokhara_fix \
  RUN_PHOKHARA/submit.sh <run> <n_seeds> <nges> 10000 <walltime>
```